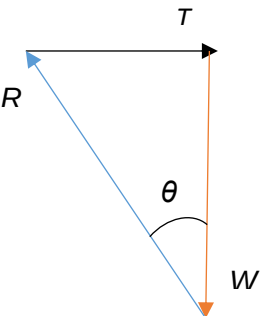


2	(a)	Principle of moments states that for an object to be in rotational equilibrium, the sum of clockwise moments about any axis must be equal to the sum of anticlockwise moments about that same axis.	
	(b)	(i)	Taking moments about the hinge, $T(1.15) = W(0.80)$ $T = \frac{5.0 \times 9.81 \times 0.80}{1.15} \quad [\text{M1}]$ $= 34.1 \text{ N} = 34 \text{ N} \quad [\text{A0}]$



		(ii)	 [B1: length of W is longer than length of T] [B1: Correct triangle]
		(iii)	$R^2 = W^2 + T^2$ $R = \sqrt{(5.0 \times 9.81)^2 + 34^2} \quad [\text{C1}]$ $= 59.7 \text{ N} \quad [\text{A1}]$ $\theta = \tan^{-1}\left(\frac{34}{5.0 \times 9.81}\right)$ $= 34.7^\circ \text{ as shown in diagram } [\text{A1}]$
		(c)	Along the slope: Gravitational potential energy of mass A is converted to kinetic energy. [B1] Along the floor: Some of the kinetic energy of mass A is converted to heat or sound as it collides with mass B and also to kinetic energy of mass B. [B1] Along the floor (until the spring is compressed at its maximum): Remaining KE (of both masses A and B) is fully converted to elastic potential energy of the spring as it is being compressed to the maximum. [B1]

3

3	(a)	1	1
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